VISHWAS BM

Robotics and AI Engineering Student

Summary

Robotics and AI Engineering student (CGPA 7.98) who builds working systems across robotics simulation, AI-assisted tools, and full-stack web and mobile apps. Built V-M.I.T.R.A, a browser-based ROS lab with AI-assisted debugging, and an autonomous trading bot (2nd prize, MANTHAN 2026). 2nd-place winner at a state-level symposium and a hackathon; technical team head for a 36-hour hackathon. Seeking a robotics or AI engineering internship.

Contact

+91 9482480642
vishwasvishu376@gmail.com
linkedin.com/in/vishwas-b-m
github.com/vishwasvishubm
Shivamogga, Karnataka

Education

JNN College of Engineering
B.E. Robotics and AI Engineering
2023 - 2027 (Expected) | CGPA 7.98

PACE U.M.P.U College
Pre-University (PCMC)
2023 | 81.17%

Sri Ramakrishna Vidyanekethana
State Board (SSLC) | 89%

Skills

- Languages: Python, C, C++, JavaScript
- AI: Gemini API, OpenAI API, prompt design, structured outputs, Claude Code
- Robotics: ROS 2, ROS Noetic and Jazzy, Gazebo, RViz, ROSBridge, SLAM, LiDAR, Isaac Sim, MuJoCo
- Web and Mobile: React.js, React Native, Expo, FastAPI, REST APIs, HTML, CSS, Tailwind CSS
- Data: pandas, SQLite, PostgreSQL, MySQL, Pydantic, PyMuPDF, data normalization
- Concepts: algorithm design, rule engines, precision, recall, F1
- Tools: Git, GitHub, Docker, Kafka, Streamlit, VS Code

Projects

V-M.I.T.R.A: ROS Lab, Team Lead (Completed)

ROS 2/Noetic, Gazebo, RViz, ROSBridge, React.js, FastAPI, PostgreSQL, Docker

- Built V-M.I.T.R.A (Virtual Mentor for Intelligent Technical and Robotics Analysis), an AI-powered browser-based ROS lab for interactive robotics learning and real-time simulation.
- Developed experiment modules on SLAM mapping, autonomous navigation, obstacle avoidance, LiDAR integration, and TurtleBot simulation.
- Added AI-assisted debugging, code analysis, and experiment guidance, plus gamification, progress tracking, and automated assessment.

Autonomous Trading Bot: 2nd Prize, MANTHAN 2026

Python, Kafka, REST APIs, pandas, Streamlit

- Built an explainable bot that trades autonomously from a live Kafka price and news stream, placing orders through a REST gateway.
- Designed a volume-flow strategy with tax planning; returned +107.9% vs the market's +81.5% on 40 unseen stocks in the official pre-flight test.
- Tested 30+ strategy variants across 32 backtest windows, with a train/test split and an independent accounting audit.
- Added auto-reconnect, crash recovery, and 96 automated tests; the pre-flight run made 4,772 API requests with zero errors.

Ghost Data Audit Agent: HackerRank Orchestrate 2026

Python, SQLite, Google Gemini API, Pydantic, PyMuPDF

- Built a privacy-compliance agent in a 24-hour hackathon that verifies customer data deletion across databases, replicas, logs, PDFs, and screenshots, with evidence-backed remediation plans for 518 requests.
- Designed a hybrid pipeline: deterministic Python for identity resolution, replica state, and policy rules, with Gemini used only for screenshot transcription and multilingual intent classification.
- Normalized 94 evidence files in 10 formats; used bipartite matching to attribute deleted records without guessing ownership.
- Reached 100% status accuracy and 96.2% step-level F1 (100% precision) with 20 adversarial tests, at about $0.08 per run.

Achievements and Leadership

- 2nd Place, Engineering a Way Out (State-Level Technical Symposium), Aditya 2K24, JNNCE Shivamogga: developed an agriculture-focused solution to improve farming methods and crop growth efficiency.
- 2nd Place, Hack-X Hackathon, MYSTERIO 6.0, JNNCE Shivamogga: built a real-time class notification system with browser and WhatsApp alerts using React.
- Technical Team Head, FOSSHACK (FOSS United 36-Hour Hackathon), PES ITM Shivamogga: led the technical side and helped conduct the event.
- Brotechkannada (@Brotechkannada): built a Kannada tech community creating AI and tech content, gadget reviews, and AI tool explainers.